WEEK 2 · FALL 2026

Machine Learning

An Overview — Part 2

From writing rules → to learning from data

Dr Muhammad Ismail

BS Computer Science · 5th Semester · Theory + Lab

RECALL

Where we stopped last week

1

Four ways to define AI

Thinking vs acting × humanly vs rationally. We chose acting rationally as the engineering target.

2

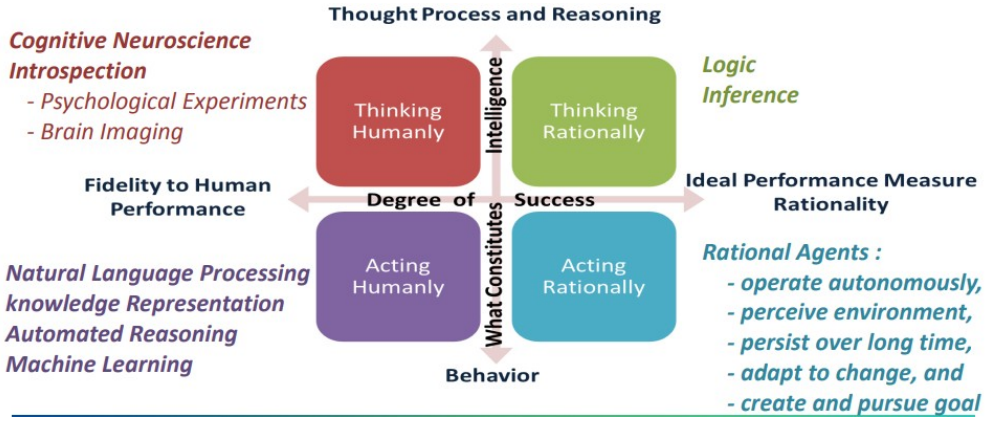
The rational agent

Perceives its environment, then acts to maximise its performance measure.

3

The unsolved part

Where do an agent's rules come from? Hand-writing them does not scale.



Week 1: the four classical definitions of AI

Today we answer the third one.

Machine learning is how an agent gets its rules — without a human writing them.

ROADMAP

The route we will take today

1

The problem

A task no one can write rules for

2

The idea

Learn the rules from examples

3

The material

Data, features, labels, notation

4

The three ways

Supervised · Unsupervised · Reinforcement

5

The reality

Where ML quietly fails

By the end you should be able to: look at any problem and say (a) is this an ML problem at all, (b) which paradigm it belongs to, (c) what data it would need.

60-SECOND CHALLENGE

Write me the rules for spam.

You are the engineer. An email arrives. Your program must output SPAM or NOT SPAM.
No machine learning allowed — only if / else. Take one minute. Write your rules.

Rule 1: if the subject contains "FREE" → spam

Rule 2: if it comes from an unknown sender → spam

Rule 3: if it has more than 3 exclamation marks → spam

Keep your list.
We will break it
on the next slide.

THE PROBLEM

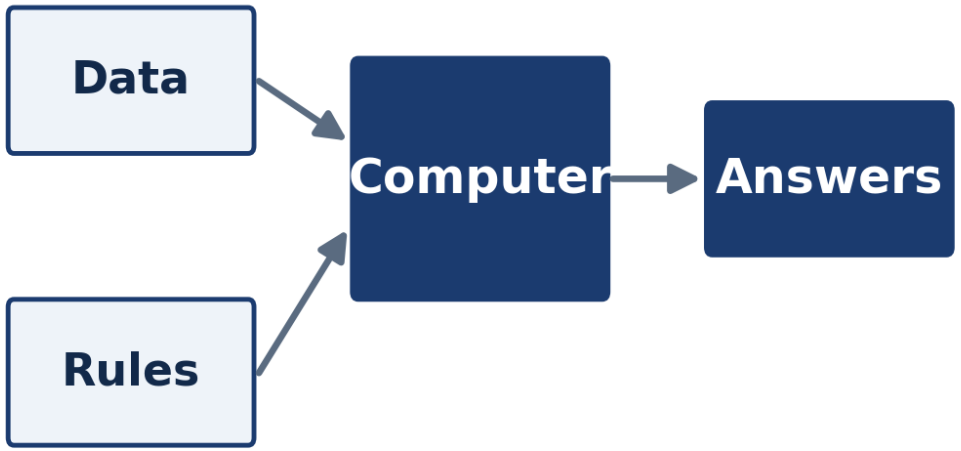
Every rule you wrote, broken in ten seconds

Contains "FREE"	"Free shuttle to campus on Friday" — from your own university.	False alarm
Unknown sender	Your admission letter. Your first job offer. Every important email you have never received before.	False alarm
Too many !!!	Spammers simply stop using them. The rule dies the day you ship it.	Adversary adapts
Add 50 more rules	They start contradicting each other. No human can hold the interactions in their head.	Unmaintainable

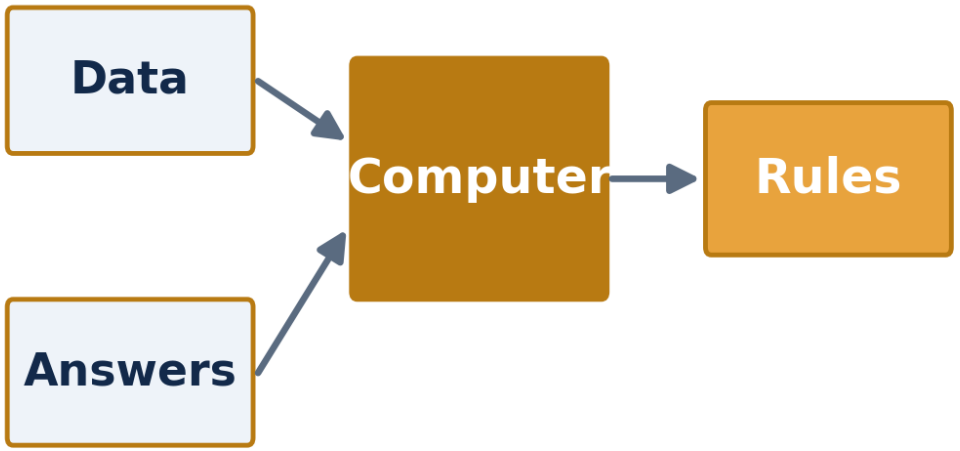
The task is real, but the rules are not writable. That is exactly the gap machine learning fills.

The shift: stop writing rules, start showing examples

Traditional Programming



Machine Learning



Traditional: the human supplies the logic; the machine executes it.

Machine learning: the human supplies examples; the machine infers the logic.

1959

The first person to make this work

> *“Field of study that gives computers the ability to learn without being explicitly programmed.”*
>
> **— Arthur Samuel, IBM, 1959**

10,000+

games the program played against itself

0

checkers strategies Samuel hand-coded

1

programmer it eventually beat — its own author

Why the checkers story still matters

- Samuel was a weak checkers player — yet his program became strong.
- The program's skill came from experience, not from his knowledge.
- So a model can exceed its creator. This is why ML is worth the trouble.
- And why we can no longer simply read the code to know what it will do.

THE ENGINEERING DEFINITION

A definition you can actually test against

> "A computer program is said to learn from experience **E** with respect to some class of tasks **T** and performance measure **P**, if its performance at tasks in **T**, as measured by **P**, improves with experience **E**." — *Tom Mitchell, 1997*

T — TASK

What the program must actually do

Classify an email as spam or not spam.

E — EXPERIENCE

The data it learns from

100,000 emails already marked by users.

P — PERFORMANCE

The number that says it improved

% of unseen emails classified correctly.

If you cannot name T, E and P, you do not yet have a machine learning problem — you have a wish.

ACTIVITY 1 · THINK – PAIR – SHARE · 4 MINUTES

Name the T, E and P.

A Predict whether a 5th-semester student will fail a course

B Recommend the next video a user should watch

C Detect a diseased leaf from a photograph of a crop

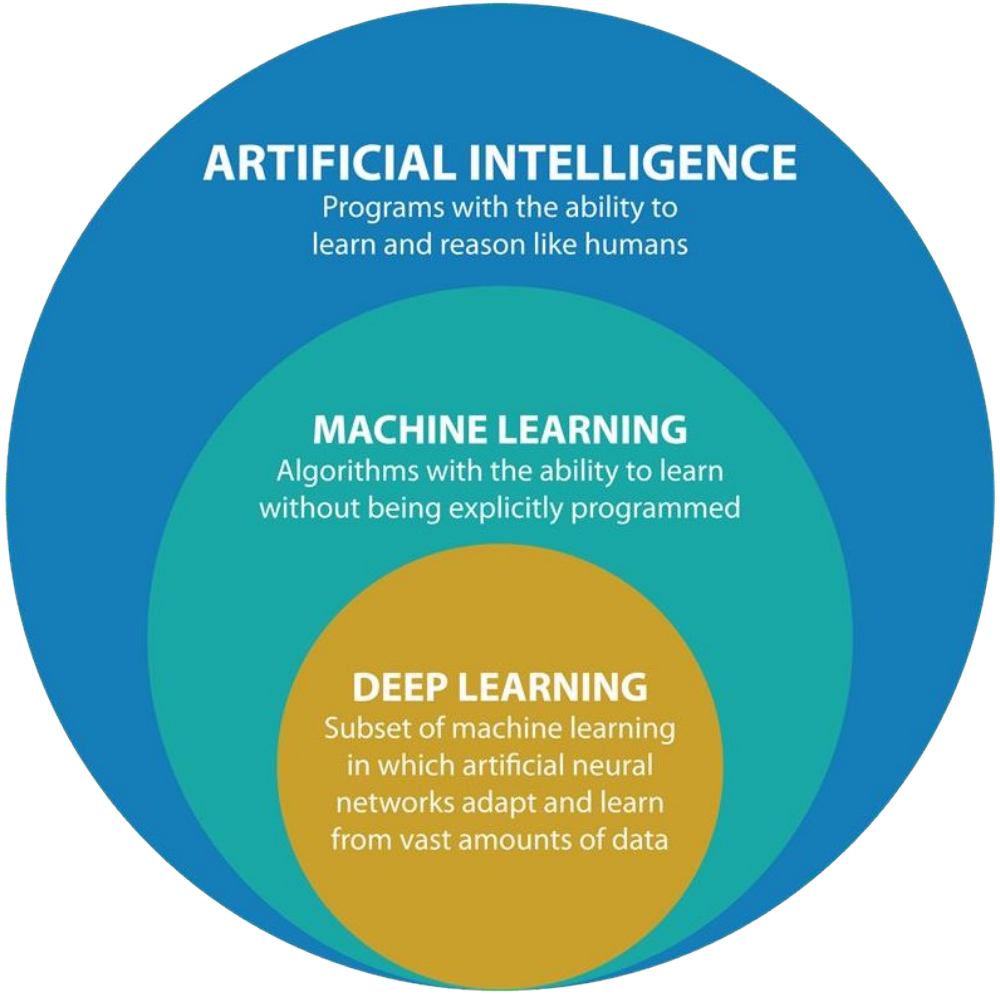
Rules

- 1 min alone — write T, E, P.
- 2 min with your neighbour — argue about P.
- 1 min: two pairs report to the class.

The hard part is never T. It is always P — and P is where projects go wrong.

RELATIONSHIP

Where machine learning sits inside AI



Artificial Intelligence

Any technique that makes a machine behave in a way we would call intelligent — including rule-based systems with no learning at all.

Machine Learning

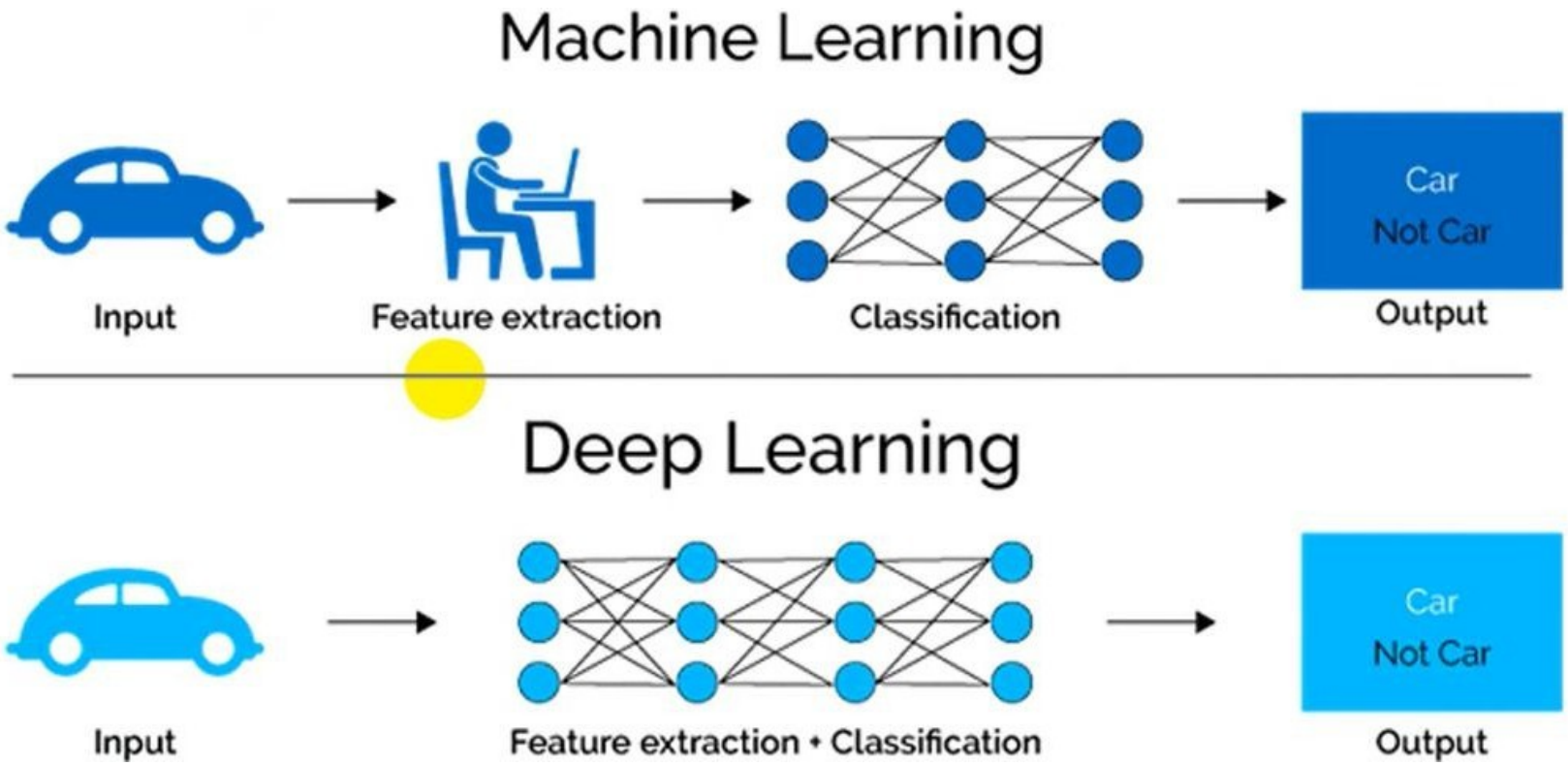
The subset that improves from data rather than from a programmer. Works well on structured, tabular data with human-chosen features.

Deep Learning

The subset of ML using many-layered neural networks. It learns the features too — which is why it dominates images, audio and text.

Nested, not parallel: every deep learning system is a machine learning system; the reverse is false.

What actually changes at the deep learning layer



Classical ML: a human decides what to measure (wheels, edges, corners). Model quality is capped by that judgement.

Deep learning: the network discovers the features itself — but it needs far more data and compute to do so.

PART 3

The raw material

Data, features, labels — and the notation we will use for the rest of the semester

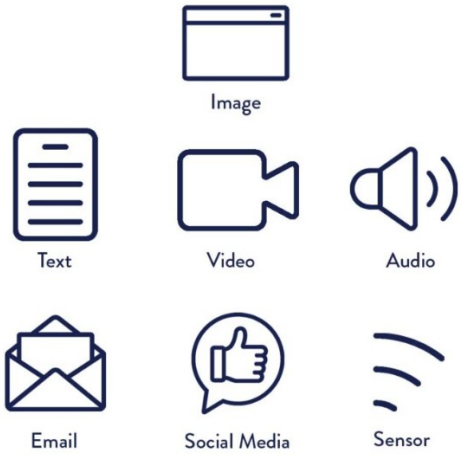
Two kinds of data — and why it matters to you

STRUCTURED · rows and columns

	Name	Type 1	Type 2	HP	Attack	Defense
2	Abra	Psychic		25	20	15
3	Kadabra	Psychic		40	35	30
4	Alakazam	Psychic		55	50	45
5	Machop	Fighting		70	80	50
6	Machoke	Fighting		80	100	70
7	Machamp	Fighting		90	130	80
8	Bellsprout	Grass	Poison	50	75	35
9	Weepinbell	Grass	Poison	65	90	50
10	Victreebel	Grass	Poison	80	105	65
11	Tentacool	Water	Poison	40	40	35
12	Tentacruel	Water	Poison	80	70	65

- Pre-defined schema; each column has a meaning and a type.
- Lives in spreadsheets and relational databases.
- Classical ML — trees, SVMs, regression — is at its best here.

UNSTRUCTURED · no fixed shape



- Images, audio, video, free text, sensor streams, social posts.
- No columns — you must create the features, or let a network learn them.
- Roughly 80% of the data an organisation holds.

TERMINOLOGY

The five words we will use every single week

	FEATURES (X)				LABEL (y)
	sepal len	sepal wid	petal len	petal wid	species
one sample →	5.1	3.5	1.4	0.2	setosa
	4.9	3.0	1.4	0.2	setosa
	7.0	3.2	4.7	1.4	versicolor
	6.3	3.3	6.0	2.5	virginica

The Iris dataset — the classic example

150 samples · 4 features · 3 classes. Small enough to hold in your head, real enough to make every mistake on. (UCI Machine Learning Repository, Fisher 1936.)

Sample / instance

one row — one flower, one email, one patient

Feature / attribute

one column of input — petal length

Label / target

the answer we want to predict — species

Training set

the rows the model is allowed to learn from

Test set

rows locked away, touched exactly once, at the end

NOTATION

The same picture, written as mathematics

The inputs

$$X \in \mathbb{R}^{150\times 4}:$$

$$\begin{bmatrix} x_1^{(1)} & x_2^{(1)} & x_3^{(1)} & x_4^{(1)} \\ x_1^{(2)} & x_2^{(2)} & x_3^{(2)} & x_4^{(2)} \\ \vdots & \vdots & \vdots & \vdots \\ x_1^{(150)} & x_2^{(150)} & x_3^{(150)} & x_4^{(150)} \end{bmatrix}$$

The answers

$$y = \begin{bmatrix} y^{(1)} \\ \dots \\ y^{(150)} \end{bmatrix} \left(y \in \{\text{Setosa, Versicolor, Virginica}\}\right)$$

Read it out loud once and it stops being frightening.

X — the design matrix — one row per sample, one column per feature

n — number of samples (rows). Here n = 150

d — number of features (columns). Here d = 4

$x^{(i)}$ — the i-th sample — a single row, a vector of d numbers

y — the label vector — one answer per row

The whole course: f(X) ≈ y

PART 4

Three ways a machine can learn

The difference between them is one question: what feedback does the learner get?

OVERVIEW

The three paradigms, in one question each

SUPERVISED

Learning with an answer key

Every training example comes with the correct answer attached.

Spam · price · diagnosis

UNSUPERVISED

Learning with no answer key

Only inputs. The algorithm must find structure that nobody labelled.

Segments · anomalies · topics

REINFORCEMENT

Learning from consequences

No answers — only rewards and penalties that arrive after acting.

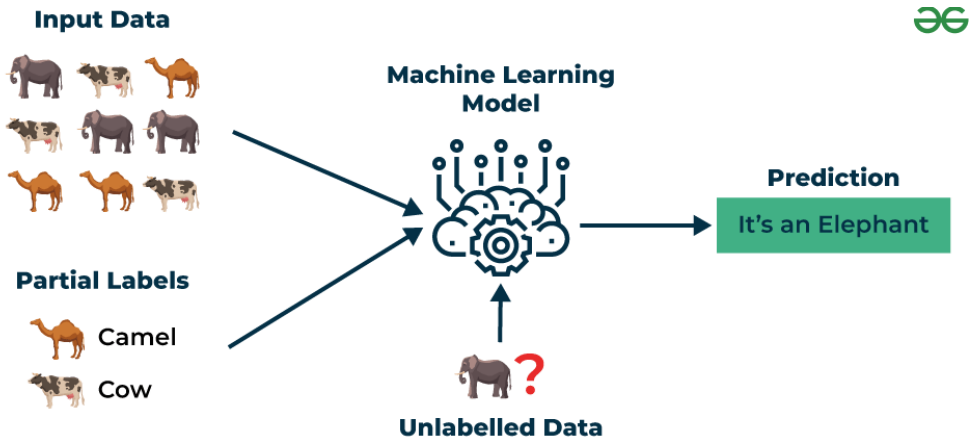
Games · robots · control

Same data can belong to different paradigms. The paradigm is set by the question you ask, not by the file you were given.

Supervised learning: someone marked the homework

Training data = Samples + Labels

- The label is the teacher. It tells the algorithm, for every example, what the right answer was.
- Learning = adjusting f until its answers match the labels on data it has already seen.
- Success = doing well on data it has NOT seen. Matching the training labels perfectly is not success; it is memorisation.
- Labelling is the expensive part — usually manual, slow, and the real bottleneck in industry.

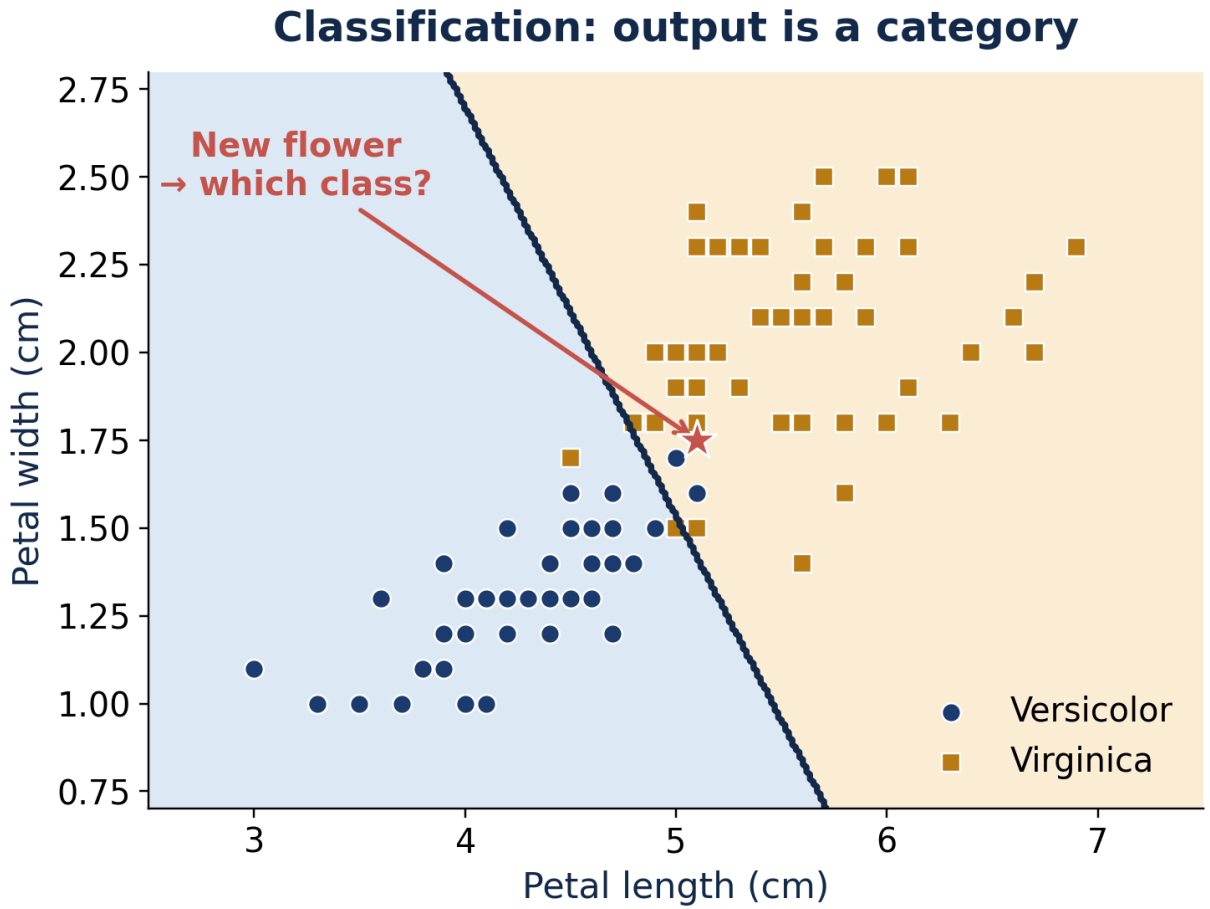


Labelled examples in, a prediction out.

Two flavours — decided entirely by the label

- **Label is a category → Classification**
- **Label is a number → Regression**

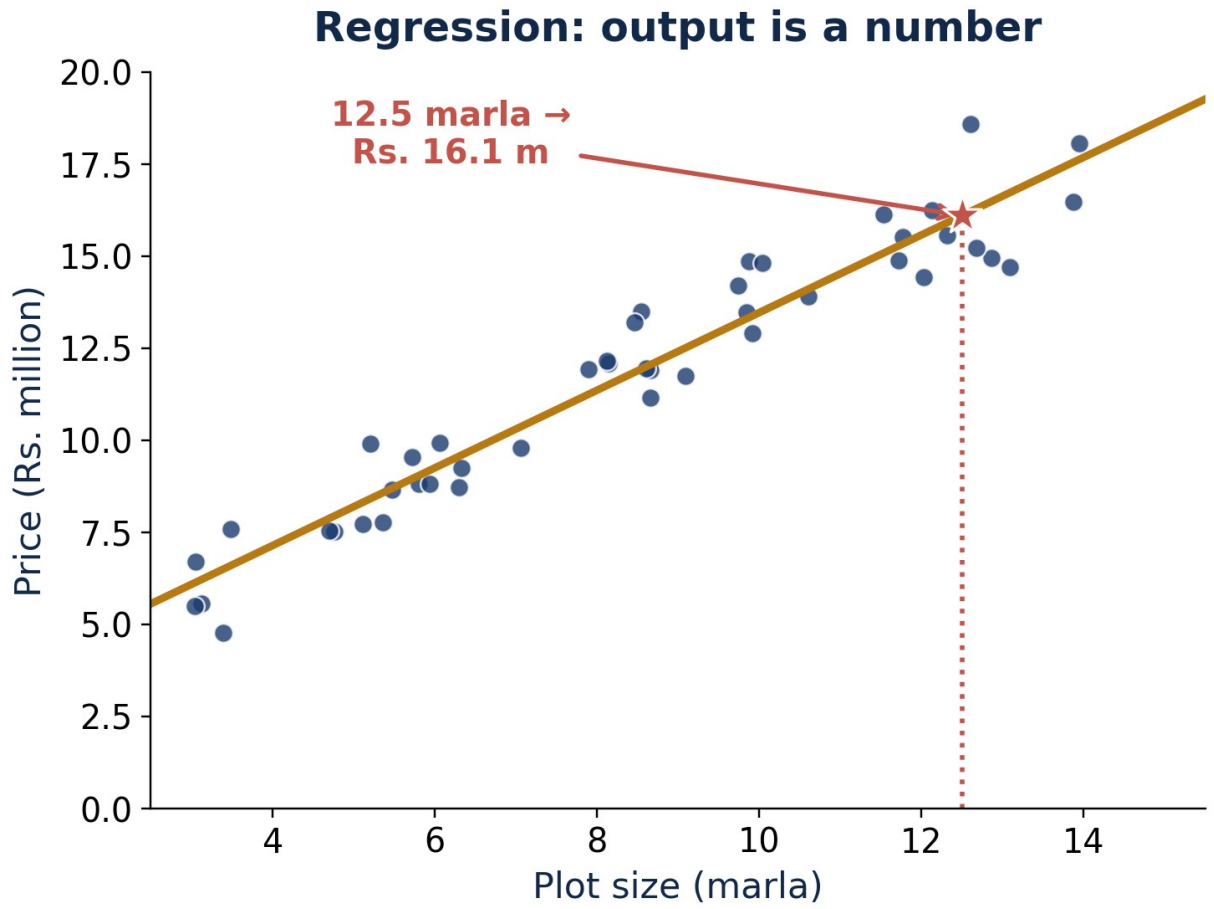
Classification: the answer is a category



- Output comes from a fixed, finite set of classes decided in advance.
- The model is really drawing a boundary; a new point is labelled by the side it falls on.
- Binary: spam / not spam. Multiclass: which of three iris species.
- Key limitation: it can only ever answer with a class it was shown. Feed it a bus, and a car-or-bike model still says car or bike.

Careful: accuracy on an imbalanced problem is misleading. 99% of emails being legitimate makes "never say spam" a 99% model.

Regression: the answer is a number

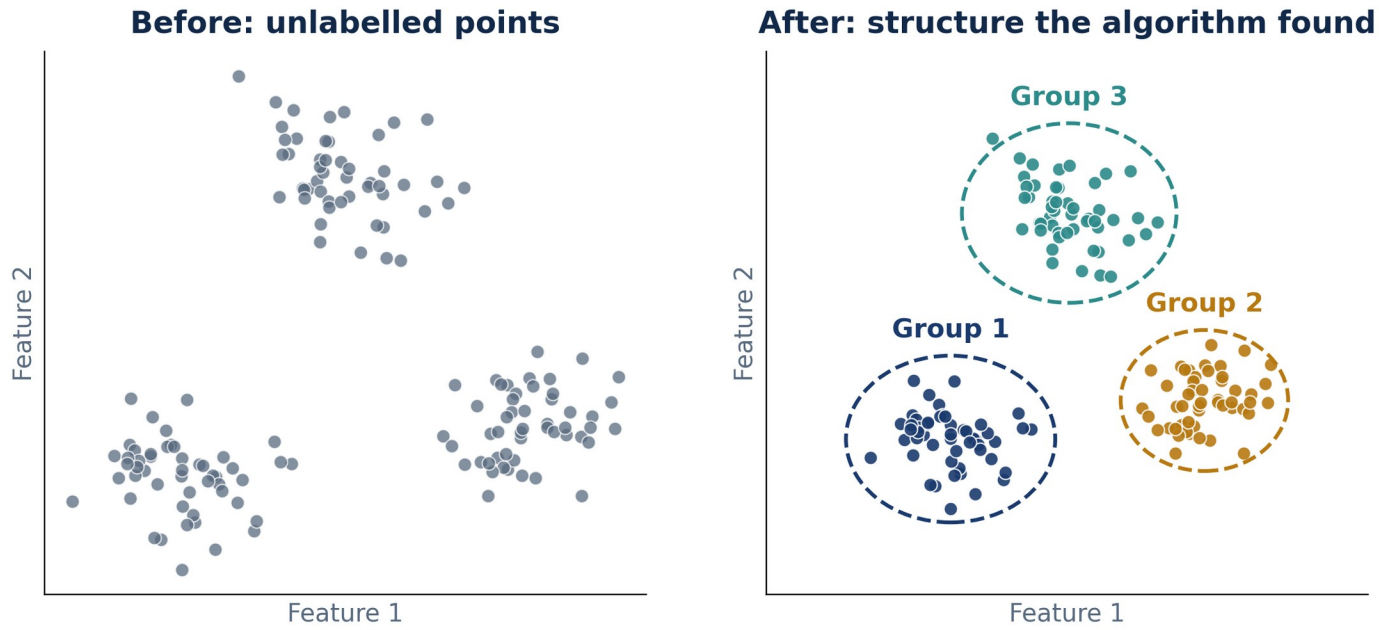


- Output is continuous — a price, a temperature, a yield, a duration.
- The model fits a curve; the prediction is the height of that curve at your input.
- There is no “right answer”, only closeness — so we measure error (MAE, RMSE) instead of accuracy.
- Predicting a number does not make it regression: predicting a house's price band (low / mid / high) is classification.

Ask yourself: is the gap between two possible answers meaningful? Rs. 5m vs Rs. 6m — yes. Cat vs dog — no.

PARADIGM 2

Unsupervised learning: no answer key exists



Clustering: the algorithm proposes the groups. Naming them is your job, not its.

Real use: a bank clusters customers, finds four spending patterns, and only then does a human look at them and say “this group is young salaried professionals”.

- Input only — no labels, no ground truth, nothing to compare against.
- Goal: expose structure that was already in the data but that nobody had written down.
- Clustering groups similar samples; dimensionality reduction compresses many features into a few; association rules find things that co-occur.
- Hard part: with no correct answer, evaluation is genuinely difficult — silhouette, elbow, stability, and human judgement.

ACTIVITY 2 · CALL IT OUT · 4 MINUTES

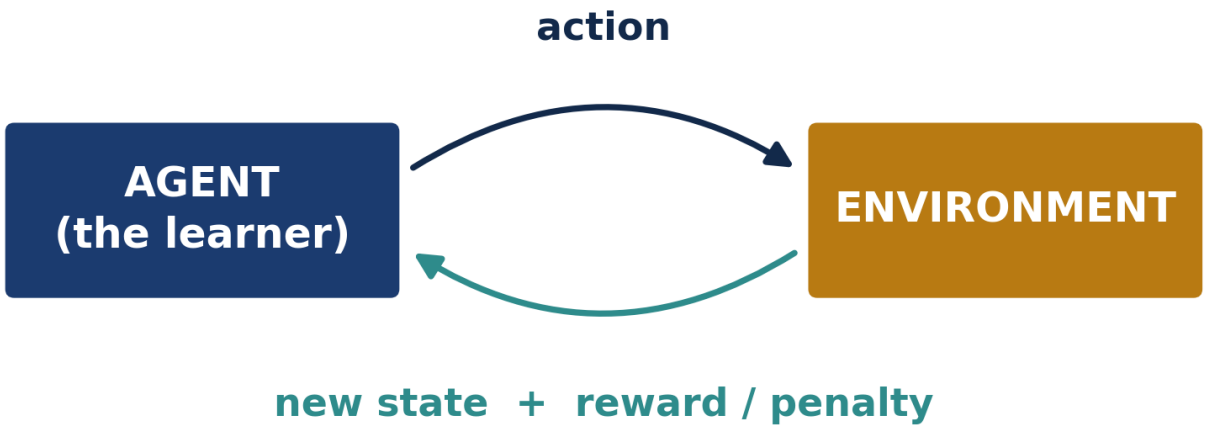
Classification, Regression, or Clustering?

1. Estimate tomorrow's electricity demand for Sukkur in megawatts.
2. Decide whether a bank transaction is fraudulent.
3. Split 50,000 customers into groups nobody has defined yet.
4. Predict a student's final CGPA from their first two semesters.
5. Route an incoming complaint to the correct department.
6. Find which products are usually bought together.

Say the paradigm, then say what the label would be. If you cannot name a label, it is not supervised.

PARADIGM 3

Reinforcement learning: learning from consequences



Goal: choose actions that maximise total reward over time

- No labelled examples. The agent acts, the environment responds, and the feedback is a reward or a penalty.
- The signal is delayed: you find out the move was bad twenty moves later. Assigning credit is the whole difficulty.
- Explore or exploit? Repeat what worked, or risk something new that might work better.
- Where it wins: games, robotics, control systems, recommendation over long sessions.

A child learning to walk gets no labelled dataset of correct steps — only the consequence of falling, and of not falling.

SUMMARY

The three paradigms, side by side

	Supervised	Unsupervised	Reinforcement
Data given	Inputs + labels	Inputs only	No dataset — an environment
Feedback	Correct answer, every time	None	Reward, usually delayed
Goal	Predict the label for new data	Discover hidden structure	Choose actions that maximise reward
Main tasks	Classification, Regression	Clustering, Dim. reduction, Association	Policy learning, Control
Judged by	Accuracy, F1, RMSE	Silhouette, stability, human review	Total reward earned
Typical use	Spam, credit risk, diagnosis	Segmentation, anomaly detection	Games, robotics

If you can fill this table from memory, you can classify any problem someone brings you.

PART 5

The part most courses skip

Where machine learning quietly fails — and the process that keeps you honest

REALITY CHECK

Four ways machine learning quietly fails

1 Garbage in, garbage out

A model cannot be better than the data behind it. Biased data produces a biased model that looks perfectly confident.

2 Correlation is not causation

Ice cream sales predict drownings. Both are caused by summer. A model that predicts is not a model that explains.

3 Memorising is not learning

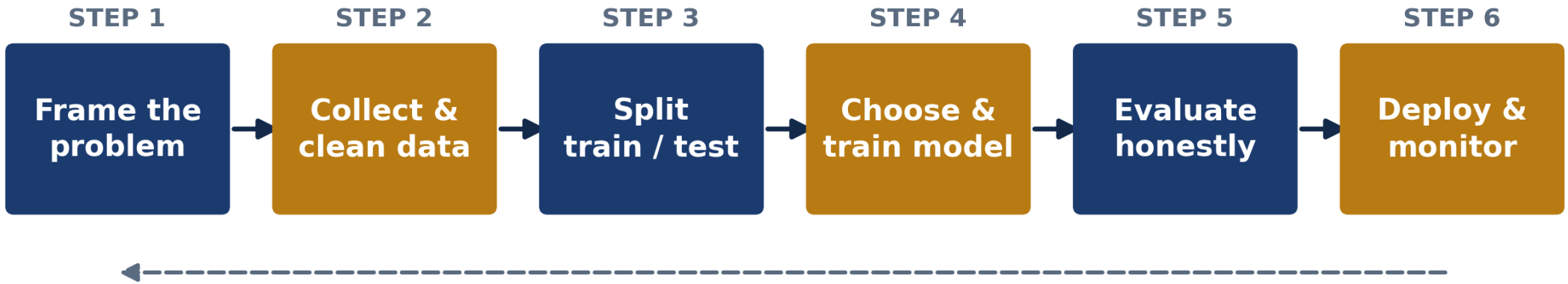
Perfect scores on training data usually mean the model memorised it and will collapse on anything new.

4 Data leakage

A feature that would not exist at prediction time inflates your results. You find out only after deployment — or in the viva.

THE PROCESS

So what does doing machine learning actually look like?



iterate — it is never a straight line

Most of the time goes here

Steps 2 and 3. Practitioners routinely spend 60–80% of a project on collecting, cleaning and splitting data.

The step everyone rushes

Step 5. An honest evaluation on untouched test data is what separates a result from a claim.

The step students forget

Step 6. A model in a notebook has helped nobody. Your project demands a working live demo.

RECAP QUIZ

Quick check — questions 1 to 3

1

A hospital wants to predict how many days a patient will stay. Which task is this, and why?

Regression — the answer is a continuous number, and the gap between 3 days and 8 days is meaningful.

2

Your model is 97% accurate at detecting a disease that affects 2% of patients. Are you happy?

No. Predicting "healthy" every time already scores 98%. Accuracy is the wrong measure here.

3

You have 10,000 customer records with no labels. Name one supervised and one unsupervised question.

Supervised needs a label you can define (will they churn?). Unsupervised does not (what natural segments exist?).

Ask the question. Wait. Then click to reveal — or hide the amber strip and let them argue first.

RECAP QUIZ

Quick check — questions 4 to 6

4 **A spam filter is retrained every night on yesterday's user reports. Name T, E and P.**

T = label an email spam / not spam. E = yesterday's user-marked emails. P = error rate on tomorrow's unseen mail.

5 **Your model scores 100% on the training set and 61% on the test set. What has happened?**

It memorised rather than learned — overfitting. Success is measured on unseen data only.

6 **You predict loan default and one feature is “amount already written off”. What is wrong?**

Data leakage — that value does not exist at the moment of prediction. The result is inflated and worthless.

Questions 5 and 6 are the two mistakes that cost the most marks in the semester project.

TAKEAWAYS

Five things to carry out of this room

1. Machine learning exists because for many real tasks, nobody can write the rules — but plenty of people can supply examples.
2. If you cannot state the Task, the Experience and the Performance measure, you do not have an ML problem yet.
3. Data is the material. Rows are samples, columns are features, the answer column is the label.
4. Three paradigms, one distinguishing question: what feedback does the learner get — an answer, nothing, or a reward?
5. The test set is touched once, at the end. Everything else in this course rests on that discipline.

NEXT WEEK

Week 3 — Your first working model

In theory

The end-to-end workflow in detail; train / validation / test; what a baseline is and why you must beat one.

In lab

scikit-learn on a real dataset: load, split, fit, predict, score — in under forty lines.

Bring with you

One candidate problem for your semester project, written as T, E and P, with a dataset URL.

Reading

- Géron, Hands-On ML — Ch. 1
- Müller & Guido, Intro to ML with Python — Ch. 1
- Mitchell (1997), Ch. 1 — the T/E/P definition
- UCI repository — browse three datasets

Questions? Office hours, or the course forum on the LMS — post it there so the whole class benefits.